1. Classification of Elements

1.1 Based on shape function order

- Simplex element: the displacement polynomial has constant and linear terms only (2-node bar, 3-node triangle, 4-node tetrahedron).
- Complex element: polynomial has higher order terms in addition to the linear terms (e.g. 6-node triangle, 8-node quad).
- Multiplex element: the boundaries are parallel to the coordinate axes and the polynomial has higher terms (e.g. rectangle with 4 nodes).

1.2 Regular and irregular elements

- Regular element: the shape is geometrically simple and symmetric – equal sides and angles (equilateral triangle, square, cube). Easy to formulate.
- Irregular element: the sides are of unequal length and the angles are arbitrary (general triangle, general quadrilateral, skewed brick). Needed to fit actual structures.
- Irregular elements are handled using natural coordinates / isoparametric mapping.
- Distorted (very skewed) elements reduce the accuracy – keep aspect ratio near 1 and angles between 45° and 135°.

1.3 2-D and 3-D elements

- 2-D: triangle (CST 3-node, LST 6-node), rectangle / quadrilateral (4, 8, 9 nodes).
- 3-D: tetrahedron (4 or 10 nodes), hexahedron or brick (8 or 20 nodes), prism / wedge (6 or 15 nodes).
- Higher order nodes lie on the mid-side of edges, giving curved boundaries and better accuracy.

2. Lagrangian and Serendipity Elements

2.1 Lagrange family

- Based on Lagrange interpolation polynomials. The element has nodes at the corners, mid-sides AND an interior node (for higher order).
- The shape function is the product of 1-D Lagrange polynomials in ξ and η:

$$N_i(\xi,\eta) = l_j(\xi) \cdot l_k(\eta)$$

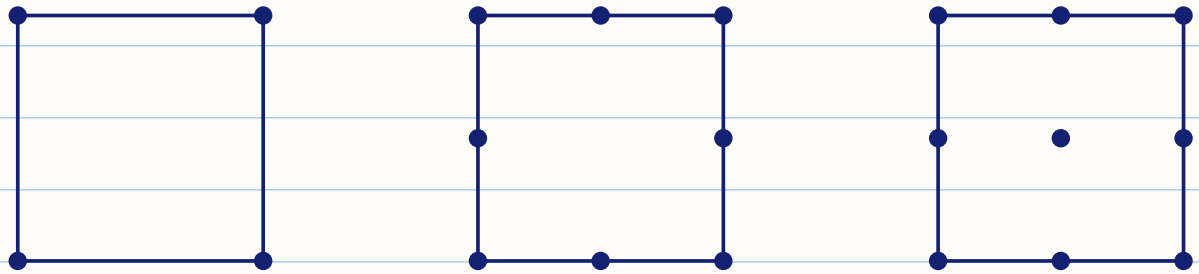
- 4-node (bilinear), 9-node (biquadratic), 16-node (bicubic) rectangles are Lagrange elements.
- Number of nodes = $(n+1)^2$ for polynomial order n.
- Disadvantage: interior nodes are extra unknowns and do not help connect neighbouring elements – they increase the size of the problem.

2.2 Serendipity family

- "Serendipity" means a lucky discovery. Nodes are only on the boundary (corners and mid-sides), NO interior node.
- Shape functions are not obtained by simple products; they are built by inspection and checked: each $N_i = 1$ at node i, 0 at others.
- 4-node, 8-node (quadratic) and 12-node (cubic) elements. Fewer nodes than Lagrange for same order, so cheaper.
- The polynomial of an 8-node serendipity element misses the x^2y^2 term:

$$u = a_1 + a_2\xi + a_3\eta + a_4\xi^2 + a_5\xi\eta + a_6\eta^2 + a_7\xi^2\eta + a_8\xi\eta^2$$

Serendipity: no interior node Lagrange: has interior node



4-node bilinear 8-node serendipity 9-node Lagrange

2.3 Comparison

- Lagrange: more nodes (9 vs 8), complete polynomial, interior node, higher cost.
- Serendipity: fewer nodes, no interior node, more economical, good for curved boundaries.
- Both give the same order of accuracy for the same polynomial order,

so serendipity is preferred in practice.

2.4 Shape functions of 4-node rectangle (Lagrange)

- With natural coordinates ξ, η from -1 to +1 and node numbering anticlockwise from lower-left:

$$N_1 = \tfrac{1}{4}(1-\xi)(1-\eta),\ N_2 = \tfrac{1}{4}(1+\xi)(1-\eta)$$
$$N_3 = \tfrac{1}{4}(1+\xi)(1+\eta),\ N_4 = \tfrac{1}{4}(1-\xi)(1+\eta)$$
$$N_1+N_2+N_3+N_4 = 1$$

2.5 Shape functions of 8-node serendipity element

- Corner nodes ($\xi_i, \eta_i = \pm 1$):

$$N_i = \tfrac{1}{4}(1+\xi\xi_i)(1+\eta\eta_i)(\xi\xi_i + \eta\eta_i - 1)$$

- Mid-side nodes with $\xi_i = 0$:

$$N_i = \tfrac{1}{2}(1-\xi^2)(1+\eta\eta_i)$$

- Mid-side nodes with $\eta_i = 0$:

$$N_i = \tfrac{1}{2}(1+\xi\xi_i)(1-\eta^2)$$

2.6 Shape functions of 9-node Lagrange element

Corner: $N_i = \tfrac{1}{4}\xi\xi_i(1+\xi\xi_i)\,\eta\eta_i(1+\eta\eta_i)$

Mid-side: $N_i = \tfrac{1}{2}(1-\xi^2)\,\eta\eta_i(1+\eta\eta_i)$ or $\tfrac{1}{2}\xi\xi_i(1+\xi\xi_i)(1-\eta^2)$

Centre: $N_9 = (1-\xi^2)(1-\eta^2)$

3. Hermite Polynomials

3.1 Concept

- In Lagrange / serendipity elements only the function value (displacement) is a nodal unknown – this gives C^0 continuity (the displacement is continuous but its slope is not).
- Hermite interpolation uses BOTH the value and its first derivative (slope) at each node, giving C^1 continuity.
- Needed for beams, plates and shells where slope must also be continuous.
- For an element with 2 nodes and 2 unknowns per node $(v, \theta) \rightarrow 4$ conditions → cubic polynomial:

$$v(x) = a_1 + a_2x + a_3x^2 + a_4x^3$$

3.2 Hermite shape functions ($s = x / l$)

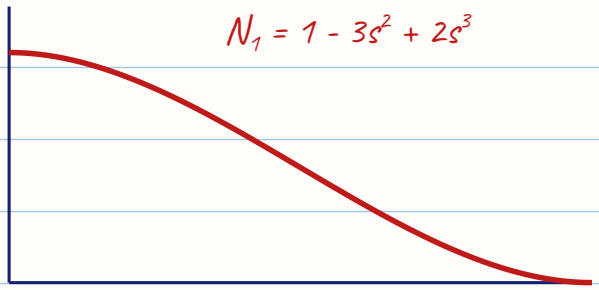
$$H_1 = 1 - 3s^2 + 2s^3$$

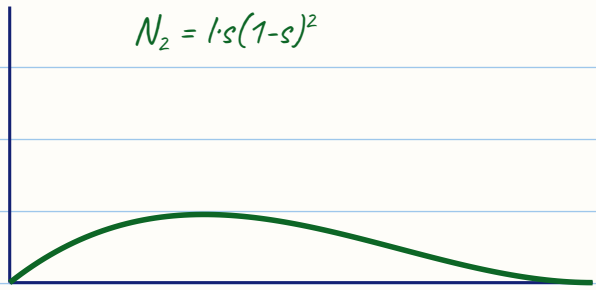
$$H_2 = l\,(s - 2s^2 + s^3)$$

$$H_3 = 3s^2 - 2s^3$$

$$H_4 = l\,(-s^2 + s^3)$$

- Conditions: $H_1(0)=1$, $H_3(l)=1$ give the displacement; $H_2'(0)=1$, $H_4'(l)=1$ give the slope.





$s = x / l$ (Hermite cubics)

4. Shape Functions up to Quadratic Formulation

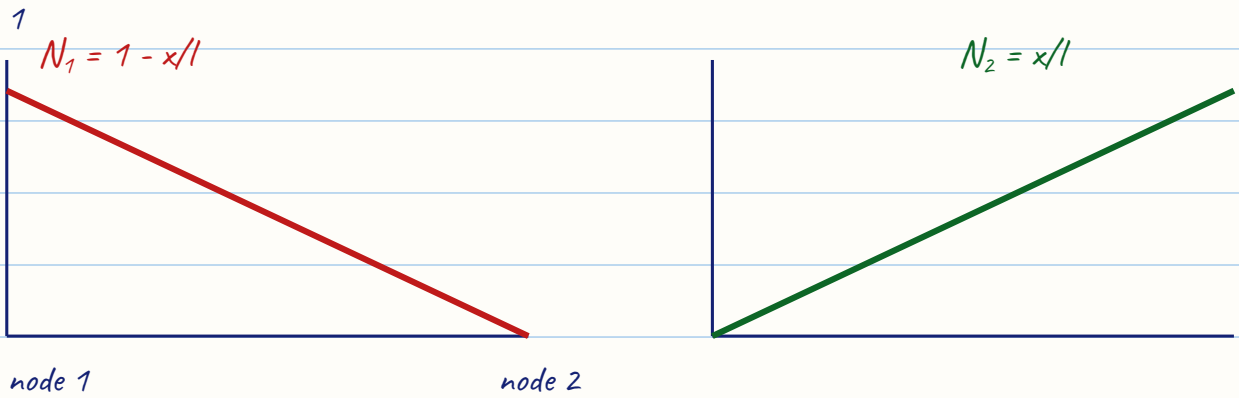
4.1 Linear 1-D element (2 nodes)

- Assume $u(x) = a_1 + a_2 x$. Apply at nodes: $u(0) = u_1$, $u(l) = u_2$.

$a_1 = u_1$, $a_2 = (u_2 - u_1) / l$

$u = (1 - x/l)\, u_1 + (x/l)\, u_2 = N_1 u_1 + N_2 u_2$

$N_1 = 1 - x/l$, $N_2 = x/l$

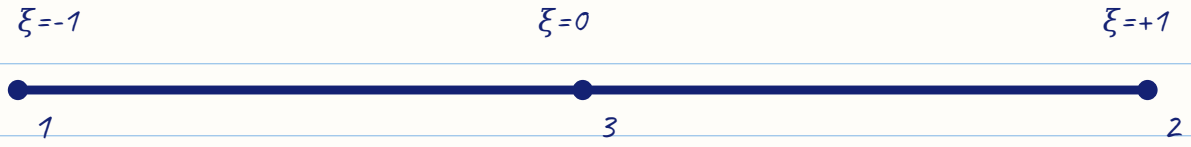


4.2 Quadratic 1-D element (3 nodes)

- Assume $u(\xi) = a_1 + a_2\xi + a_3\xi^2$ with nodes at $\xi = -1, +1$ and 0 (mid-node).

  $N_1 = \frac{1}{2}\xi(\xi - 1)$, $N_2 = \frac{1}{2}\xi(\xi + 1)$, $N_3 = 1 - \xi^2$

- Check: at $\xi = -1$, $N_1 = 1$; at $\xi = 1$, $N_2 = 1$; at $\xi = 0$, $N_3 = 1$; sum = 1.



$N_1 = \xi(\xi-1)/2$, $N_2 = \xi(\xi+1)/2$, $N_3 = 1-\xi^2$

3-node (quadratic) bar element

4.3 Linear triangle (3 nodes)

- $u = a_1 + a_2x + a_3y$. Solving with nodal values gives

  $N_i = (a_i + b_ix + c_iy) / 2A$, $i = 1, 2, 3$

  $a_1 = x_2y_3 - x_3y_2$, $b_1 = y_2 - y_3$, $c_1 = x_3 - x_2$

- (others by cyclic permutation $1 \to 2 \to 3 \to 1$) and $2A = x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)$.

4.4 Quadratic triangle (6 nodes) in area coordinates

$$N_1 = L_1(2L_1 - 1),\ N_2 = L_2(2L_2 - 1),\ N_3 = L_3(2L_3 - 1)$$

$$N_4 = 4L_1L_2,\ N_5 = 4L_2L_3,\ N_6 = 4L_3L_1$$

- Nodes 1-3 are corners, nodes 4-6 are mid-sides.

4.5 Linear tetrahedron (4 nodes)

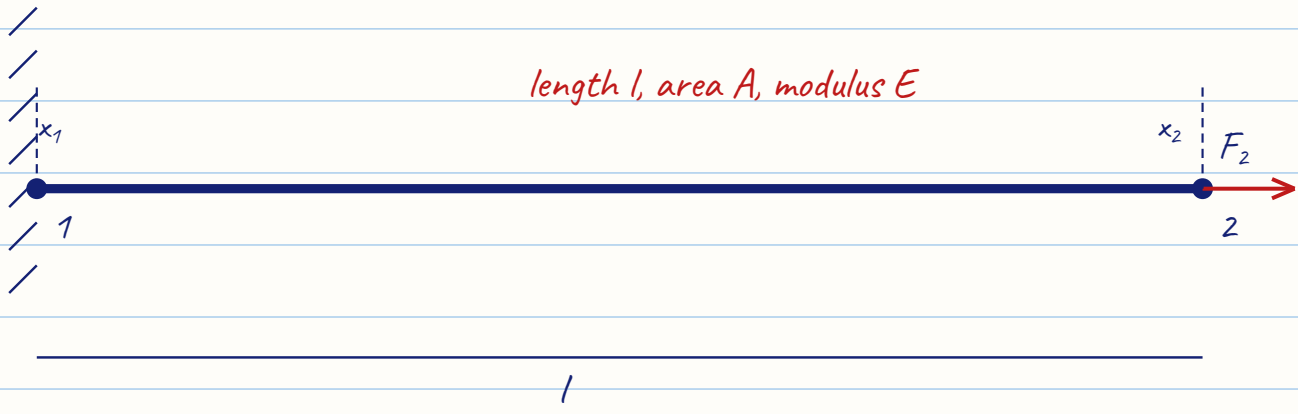
$$N_i = (a_i + b_ix + c_iy + d_iz) / 6V$$

- In volume coordinates $N_i = L_i$, with $L_1+L_2+L_3+L_4 = 1$.

5. FEA of One-Dimensional Problems – Bar Element

5.1 Bar element

- A bar (rod) carries only axial load. Each node has one dof (axial displacement u).
- Let length = l, cross-sectional area = A, Young's modulus = E.



$$u(x) = N_1u_1 + N_2u_2$$

5.2 Displacement function and shape functions

$$u = N_1 u_1 + N_2 u_2 = [N]\{q\},\ [N] = [\,1 - x/l \quad x/l\,]$$

$$\{q\} = \{u_1\ u_2\}^T$$

5.3 Strain – displacement relation

$$\varepsilon = du/dx = [\,-1/l \quad 1/l\,]\{u_1\ u_2\}^T = [B]\{q\}$$

$$[B] = (1/l)[\,-1 \quad 1\,]$$

- Strain in a linear bar element is constant.

5.4 Stress – strain relation

$$\sigma = E\varepsilon = E[B]\{q\} = (E/l)(u_2 - u_1)$$

- Axial force in the bar $P = \sigma A = (AE/l)(u_2 - u_1)$.

5.5 Stiffness matrix by direct method (from Hooke's law)

- Force at node 2: $F_2 = (AE/l)(u_2 - u_1)$; for equilibrium $F_1 = -F_2$.

$$\{F_1\ F_2\}^T = (AE/l)[\,1 \ -1\ ;\ -1 \ \ 1\,]\{u_1\ u_2\}^T$$

$$[k] = (AE/l)[\,1 \ -1\ ;\ -1 \ \ 1\,]$$

5.6 Stiffness matrix by energy method

- Strain energy of the element:

$$U = \tfrac{1}{2}\int \sigma\,\varepsilon\,A\,dx = \tfrac{1}{2}\{q\}^T\left[\int [B]^T E [B] A\,dx\right]\{q\}$$

$$[k] = \int_0^l [B]^T E [B] A\,dx = AEl[B]^T[B]$$

$$[k] = (AE/l)[\,1 \ -1\ ;\ -1 \ \ 1\,]$$

- The matrix is symmetric, singular (determinant = 0) because rigid

body motion is not yet prevented, and has diagonal terms always positive.

5.7 Load vectors

- Body force f per unit volume (e.g. self weight, ρg):

  $\{f_b\} = A f l / 2 \{1\ 1\}^T$

- Traction / distributed load T per unit length:

  $\{f_T\} = T l / 2 \{1\ 1\}^T$

- Point loads are applied directly at the nodes.

5.8 Temperature effects

- Thermal strain $\varepsilon_0 = \alpha \Delta T$. Equivalent nodal force:

  $\{f_{th}\} = A E \alpha \Delta T \{-1\ 1\}^T$

5.9 Bar of variable cross-section

- Use the mean area $A = (A_1 + A_2) / 2$ for a tapered bar, or divide into several elements of constant area.

  $[k] = (E (A_1+A_2) / 2l) [\,1\ -1\ ;\ -1\ 1\,]$

5.10 Quadratic (3-node) bar element

- Nodes: 1 at $\xi = -1$, 2 at $\xi = +1$, 3 at $\xi = 0$; and $dx = (l/2)\, d\xi$.

  $[B] = (2/l) [\,\xi - ½\ \xi + ½\ -2\xi\,]$

  $[k] = (AE / 3l) [\,7\ 1\ -8\ ;\ 1\ 7\ -8\ ;\ -8\ -8\ 16\,]$

6. Assembly of Bar Elements – Procedure

- Step 1: Divide the bar into elements; number the nodes and elements.
- Step 2: Write [k] of each element in terms of local dof.
- Step 3: Assemble into the global stiffness matrix [K] by adding terms at common dof.

  $$[K] = \Sigma\, [k^{(e)}]$$
- Step 4: Global equation $[K]\{U\} = \{F\}$.
- Step 5: Apply boundary conditions ($u = 0$ at the fixed support) by the elimination method or the penalty method.
- Step 6: Solve for nodal displacements U.
- Step 7: Find element stresses $\sigma = E\,[B]\{q\}$ and support reactions $R = K\,U - F$.

6.1 Illustration – two element stepped bar

- Two elements (1-2) and (2-3) with $(AE/l)_1 = k_1$, $(AE/l)_2 = k_2$.

  $$[K] = [\, k_1 \; -k_1 \; 0 \; ; \; -k_1 \; k_1+k_2 \; -k_2 \; ; \; 0 \; -k_2 \; k_2 \,]$$
- Fixed at node 1: $U_1 = 0$, load P at node 3 gives

  $$U_2 = P/k_1,\; U_3 = P/k_1 + P/k_2$$
- Stresses: $\sigma_1 = E_1(U_2 - U_1)/l_1$, $\sigma_2 = E_2(U_3 - U_2)/l_2$.

6.2 Elimination and penalty methods

- Elimination: strike off the row and column of the fixed dof, solve the remaining equations.
- Penalty approach: add a very large number C ($C \approx 10^4 \times \max K_{ij}$) to

the diagonal term of the fixed dof, and C·a to the force vector (a = prescribed displacement).

$$(K_{ii} + C)\, U_i = F_i + C\, a$$

7. Typical Problem (Solved Example)

- Problem: A steel bar of length 400 mm is fixed at one end and has two segments: segment 1, $A = 600\ mm^2$, $l = 200$ mm; segment 2, $A = 400\ mm^2$, $l = 200$ mm. $E = 200$ GPa. A load of 20 kN acts at the free end. Find displacements and stresses.

$k_1 = A_1E/l_1 = 600 \times 200000 / 200 = 6 \times 10^5$ N/mm

$k_2 = A_2E/l_2 = 400 \times 200000 / 200 = 4 \times 10^5$ N/mm

After applying $U_1 = 0$: $[k_1+k_2\ -k_2\ ;\ -k_2\ k_2\ ]\ \{U_2\ U_3\} = \{0\ 20000\}$

$U_2 = 20000 / 6\times10^5 = 0.0333$ mm, $U_3 = 0.0333 + 20000/4\times10^5 = 0.0833$ mm

$\sigma_1 = E\,(U_2 - 0)/l = 200000 \times 0.0333/200 = 33.3\ N/mm^2$

$\sigma_2 = 200000 \times (0.0833 - 0.0333)/200 = 50\ N/mm^2$

- Check: $\sigma_1 = P/A_1 = 20000/600 = 33.3$ □. $\sigma_2 = 20000/400 = 50$ □.

Exam tip: derive [k] for the bar (both methods), shape functions, assembly and one numerical problem are the usual long questions.